

LAB 211 Assignment

Type: Long Assignment
Code: J1.L.P0037
LOC: 450
Slot(s): N/A

Title

Employee Payroll Management System.

Background

- A medium-sized company requires a **Java console-based application** to manage employee information and calculate monthly payroll. Each employee has personal details, job-related information, and salary data.
- The system must store employee records in **text files**, support **salary calculation**, and allow **searching, updating, and reporting payroll information**.
- File description

File employees.txt	Description
E004, David Miller, Developer, 1300, 21, 250, active E005, Emma Wilson, Tester, 950, 20, 100, active E006, Frank Brown, HR, 1100, 22, 150, active E007, Grace Taylor, Developer, 1250, 24, 300, active E008, Henry Anderson, Manager, 2200, 19, 600, active E009, Ivy Thomas, Tester, 980, 18, 120, inactive E010, Jack Moore, Developer, 1400, 26, 350, active E011, Karen Lee, HR, 1050, 20, 80, active E012, Leo Martin, Manager, 2100, 17, 500, inactive E013, Nina Clark, Developer, 1350, 23, 280, active	Each line represents one employee record in the following format: <id, name, role, baseSalary, workingDays, bonus, status> (You need to be aware that the data reading process may encounter lines containing erroneous data.)

Constraints

1. Employee ID

- Cannot be empty
- Must be **unique**
- Format: **E** followed by **3 digits** (e.g., E001)

2. Employee Name

- Cannot be empty

3. Role

- Must be one of the following positions: **Developer – Tester – Manager – HR**

4. Base Salary

- Must be a **positive number**

5. Working Days

- Integer value
- Range: **0 – 26**

6. Bonus

- Must be ≥ 0

7. Status

- Only one of the two values describing the working state, **active** or **inactive**, is accepted

Program Specifications

Students are required to build a **Java console application** using the **OOP approach** to manage employee information and calculate monthly payroll.

The program must implement the following functions:

- 1- Load employee data from file
- 2- Add a new employee
- 3- Update employee information
- 4- Remove an employee by ID
- 5- Search employees by attribute
- 6- Calculate monthly payroll
- 7- Display employee list
- 8- Save data to file
- 9- Quit program

Features and Assessment:

No.	LOCs	Function
1/	30	Load from file: Read employee data from employees.txt.
2/	70	Add employee: Input, validate, and store new employee.
3/	70	Update employee: Update role, salary, bonus, or status.
4/	50	Remove employee: Delete employee by ID.
5/	50	Search employee: Search by ID, name, role, or status.
6/	70	Calculate payroll: Compute total salary for each active employee.
7/	60	Display Employee List: Display all employees.
8/	30	Save to file: Write updated data back to file.
9/	20	Quit program: Confirm saving before exit.

Note for Students:

- **Follow OOP principles** (encapsulation, inheritance, polymorphism, abstraction).
- **Follow Computational Thinking** (Decomposition, Pattern Recognition, Abstraction, and Algorithm) for reporting and presentation.
- **Input validation must strictly follow constraints.**
- **Show appropriate messages for invalid or missing data.**
- **Evaluation: Understanding, Correctness, data validation, and coding style.**
- **Submission**
 - **Source code**

- ***Diagram***
- ***Flow-chart***